CSE 4345: Big Data Analytics

Week 7, Part 1 — Querying Big Data: Hive, HiveQL & the SQL-on-Hadoop Ecosystem

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CSE, PUC, Spring 2026

Course & Lecture Information

Lecture Identity

Course:	CSE 4345
Week / Part:	7 of 11 | Part 1 of 2
CLO:	CLO3, CLO4
PLO:	PLO(e), PLO(f)

CLO3 & CLO4

CLO3: *Explain big data techniques and tools used to store, manage, and analyse large datasets*

CLO4: *Solve big data problems and build a project applying big data tools and frameworks*

Course Progression

Week 6: Spark, RDDs, DataFrames, lazy evaluation
Week 7: Hive, HiveQL, SQL-on-Hadoop ecosystem

Part 1 Covers

- Why SQL-on-Hadoop? The analyst accessibility gap
- SQL-on-Hadoop ecosystem: Hive, Presto, Impala, Drill, Spark SQL
- Hive architecture: HiveServer2, Driver, Metastore, execution engine
- HiveQL query compilation: parse → analyse → optimise → MapReduce/Tez/Spark
- The Hive Metastore: what it stores and why it is the control plane
- Schema-on-read vs. schema-on-write: tradeoffs and when to use each
- HiveQL vs. SQL: key differences (ACID, UPDATE, latency)

Lecture Agenda

1. The SQL-on-Hadoop Movement
2. Apache Hive Architecture
3. Schema-on-Read vs. Schema-on-Write
4. HiveQL: Syntax, Partitioning, and Bucketing
5. Presto: Interactive SQL on Hadoop
6. Student Assessment Pool

The SQL-on-Hadoop Movement

Why SQL-on-Hadoop? The Analyst Accessibility Gap

The Problem Before Hive (pre-2008)

Hadoop's power came with a steep learning curve:

- Every query had to be written as a **Java MapReduce program** — 100+ lines of boilerplate for a simple `GROUP BY`
- Business analysts, data scientists, and BI teams speak **SQL** — not Java
- Time-to-insight for a simple aggregation: **days** (write, test, debug MapReduce job)
- Facebook needed to enable thousands of non-engineers to query petabytes of user data

The Solution: Declare, Don't Program

Apache Hive (2008, Facebook) introduced a **SQL-like query language (HiveQL)** that automatically compiles into MapReduce jobs, Tez DAGs, or Spark jobs — depending on the execution engine configured. The analyst writes `SELECT`, Hive writes the distributed job.

SQL-on-Hadoop Ecosystem Overview

Tool	Year / Origin	Execution Model	Latency	Best For
Apache Hive	2008 Facebook	MR / Tez / Spark (compile-time)	Minutes	Batch analytics large ETL jobs
Presto (Trino)	2013 Meta (Facebook)	In-memory pipeline execution	Seconds	Interactive BI dashboards
Impala	2012 Cloudera	Native C++ MPP engine	Seconds	Low-latency SQL on HDFS/S3
Spark SQL	2014 Databricks	Catalyst optimizer in-memory RDDs	Seconds	Unified batch + streaming SQL
Apache Drill	2012 MapR	Schema-free JSON-native	Seconds	Exploratory SQL on JSON/Parquet

Shared Storage: HDFS | S3 | Azure ADLS | GCS

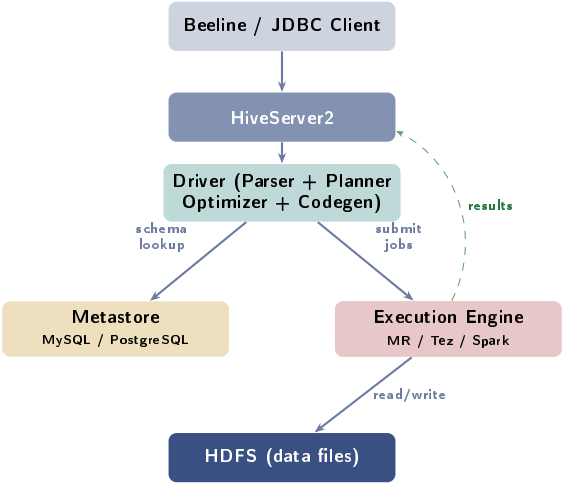
Apache Hive Architecture

Hive Architecture: From HiveQL to HDFS

Component Roles

- **HiveServer2**: JDBC/ODBC server; authenticates users; accepts HiveQL queries from Beeline CLI, BI tools (Tableau, Power BI), and JDBC drivers
- **Driver**: core query processing pipeline — Parser → Semantic Analyser → Logical Planner → Query Optimizer → Physical Plan Generator
- **Metastore**: relational database (MySQL / PostgreSQL) storing all metadata: table schemas, column types, partition information, file locations on HDFS
- **Execution Engine**: pluggable — can be **MapReduce** (default legacy), **Tez** (DAG of tasks, faster), or **Spark** (`SET hive.execution.engine=spark`)
- **HDFS**: actual data storage — Hive tables are directories; partitions are sub-directories

Hive Architecture: From HiveQL to HDFS



The Hive Metastore: The Control Plane of Your Data Lake

What the Metastore Stores

The Metastore is a **relational database** (typically MySQL or PostgreSQL) that acts as the **central catalogue** for all structured data on HDFS:

- **Database & Table definitions**: column names, data types, comment descriptions
- **Partition metadata**: which partition values exist (e.g., `year=2024, month=1`) and the HDFS path for each partition directory
- **SerDe (Serializer/Deserializer)**: how to parse the physical file format (ORC, Parquet, CSV, JSON) into column values
- **Table statistics**: row counts, column min/max, NULL counts (used by the query optimizer)
- **Privileges**: table-level and column-level access control

The Hive Metastore: The Control Plane of Your Data Lake

Why the Metastore is Critical

Without the Metastore, Hive cannot answer a single query — it needs to know *where* the data lives on HDFS and *how* to parse it. This is why the Metastore is often deployed as a **highly available service** (Metastore HA) in production clusters.

Spotify Scale

Spotify's Hive Metastore catalogs over **1 million tables** across a cluster processing 600 billion events per day. The Metastore itself runs on a replicated MySQL cluster with automatic failover. **Lesson**: The Metastore is as important as HDFS itself. Metastore corruption = inability to query any table even though data on HDFS is intact.

Schema-on-Read vs. Schema-on-Write

Schema-on-Read vs. Schema-on-Write

Property	Schema-on-Write (RDBMS)	Schema-on-Read (Hive)
When schema enforced	At INSERT / LOAD time	At SELECT / query time
Invalid data handling	Rejected immediately	Stored; NULL returned at query
Flexibility	Rigid: must define schema upfront	Flexible: schema applied to any file
Write speed	Slower (validation overhead)	Fast: raw file copy to HDFS
Query speed	Faster (data pre-validated)	Slower (parse & validate per query)
Multiple schemas	One schema per table	**Many schemas on same data**
Storage format	Proprietary (InnoDB pages)	Open: ORC, Parquet, CSV, JSON, Avro
Use case	OLTP: e-commerce orders, banking	OLAP: log analysis, batch reporting
Example	MySQL: `INSERT` fails if `amount` is text	Hive: stores "abc" in amount column; query returns `NULL`

Schema-on-Read vs. Schema-on-Write

Schema-on-Read Advantage

Load raw web logs immediately to HDFS without knowing the final schema. Define multiple different Hive tables over the same log files as requirements evolve. No ETL transformation needed at ingest time.

Schema-on-Read Risk

Data quality issues are silent until query time. A corrupted column returns NULL rather than an error. Production data pipelines should combine schema-on-read Hive with downstream validation (Great Expectations, dbt tests) to catch bad data.

HiveQL: Syntax, Partitioning, and Bucketing

HiveQL vs. SQL: Key Differences

Feature	SQL (RDBMS)	HiveQL
Schema enforcement	On write (strict)	On read (flexible)
ACID transactions	Full: INSERT, UPDATE, DELETE, ROLLBACK	Limited: ORC tables with ACID mode only
UPDATE / DELETE	Any row, any time	Only in ACID-enabled transactional tables
Indexing	B-tree, hash, bitmap	Partitions + Buckets (no traditional indexes)
Sub-queries	Correlated sub-queries	Non-correlated only (some limitations)
Query latency	Milliseconds	Minutes (batch MapReduce) / Seconds (Tez/Spark)
Execution model	Single-node or MPP	Distributed: MR / Tez / Spark DAG
Data location	Database storage engine	HDFS directories (user-managed)
Scale	Terabytes max	Petabytes
Suitable workload	OLTP (real-time reads/writes)	OLAP (batch analytics, reporting)

HiveQL vs. SQL: Key Differences

The Most Common Misconception

The common misconception is that HiveQL = SQL and expect millisecond query response. Hive is a **batch system**: a simple `SELECT COUNT(*)` on a 1 TB table may take 5–10 minutes on MapReduce. Use Presto or Spark SQL for interactive latency.

HiveQL: Table Creation and Basic Queries

Creating and Loading a Table

```
-- External table: data stays in HDFS
-- even if table is dropped
CREATE EXTERNAL TABLE transactions (
  txn_id   BIGINT,
  user_id  STRING,
  amount   DOUBLE,
  country  STRING,
  txn_date DATE
)
ROW FORMAT DELIMITED
  FIELDS TERMINATED BY ','
STORED AS ORC
LOCATION '/data/bkash/txns/';

-- Load data (moves file to table dir)
LOAD DATA INPATH
  '/staging/txns_2024.orc'
  INTO TABLE transactions;
```

HiveQL Aggregation Queries

```
-- Top 5 countries by total amount
SELECT   country,
         SUM(amount) AS total_BDT
FROM     transactions
WHERE    YEAR(txn_date) = 2024
GROUP BY country
ORDER BY total_BDT DESC
LIMIT    5;

-- Monthly revenue trend
SELECT   MONTH(txn_date) AS mo,
         COUNT(*)        AS num_txns,
         AVG(amount)     AS avg_amount
FROM     transactions
WHERE    country = 'BD'
  AND    YEAR(txn_date) = 2024
GROUP BY MONTH(txn_date)
ORDER BY mo;
```

Partitioning: Directory-Level Data Pruning

What Is Partitioning?

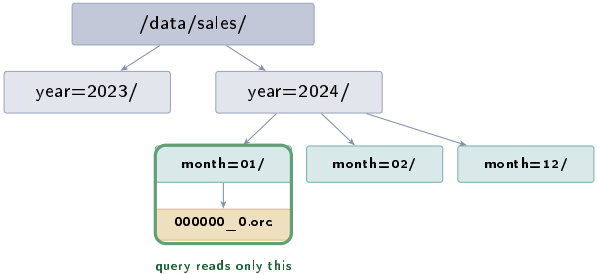
Partitioning physically organises table data into **separate HDFS sub-directories** based on the values of one or more columns. When a query filters on the partition key, Hive reads **only the matching sub-directories** — skipping all others.

- **Partition key**: typically a low-cardinality column used frequently in `WHERE` (date, country, region, status)
- **Benefit**: avoids full-table scan; reduces data read from HDFS; cuts job time proportionally
- **Example**: query on `year=2024, month=01` reads only 1 sub-directory, not 36 months of data

Partition Explosion Risk

Never partition on high-cardinality columns (e.g., `user_id`, `txn_id`). A 100 M user table would create 100 M HDFS directories — crashing the NameNode with metadata overflow.

Partitioning: Directory-Level Data Pruning



WHERE year=2024 AND month=01

skips year=2023/ and month=02..12/

Partitioning: DDL and Dynamic Partition Insert

Creating a Partitioned Table

```
CREATE TABLE sales (
  txn_id  BIGINT,
  amount  DOUBLE,
  country STRING
)
PARTITIONED BY (year INT, month INT)
STORED AS ORC;

-- Static partition load
INSERT INTO sales
  PARTITION (year=2024, month=1)
SELECT txn_id, amount, country
FROM   raw_sales
WHERE  sale_year = 2024
  AND  sale_month = 1;
```

Query Pruning in Action

```
-- Hive reads ONLY year=2024/month=1/
-- Skips all other partitions
SELECT SUM(amount)
FROM   sales
WHERE  year=2024 AND month=1;
```

Partitioning: DDL and Dynamic Partition Insert

Dynamic Partition Insert (all months at once)

```
-- Enable dynamic partitioning
SET hive.exec.dynamic.partition=true;
SET hive.exec.dynamic.partition.mode
    =nonstrict;

-- Hive infers partition values
-- from the last SELECT columns
INSERT INTO sales
  PARTITION (year, month)
SELECT txn_id,
       amount,
       country,
       YEAR(sale_date)  AS year,
       MONTH(sale_date) AS month
FROM   raw_sales;
```

Rule

Partition columns must appear **last** in the SELECT list when using dynamic partition INSERT. If you specify `PARTITION (year, month)` but select `month` before `year`, Hive silently assigns values incorrectly.

Bucketing: Hash-Based Clustering for Efficient JOINs

What Is Bucketing?

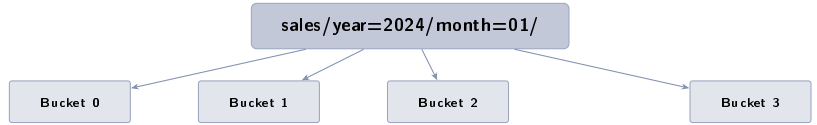
Bucketing further divides each partition into a **fixed number of files** (buckets) using a **hash function** on a chosen column. Rows with the same hash(column) land in the same bucket file.

1. Define `CLUSTERED BY (column) INTO N BUCKETS`
2. All rows where `hash(column) % N = k` are written to bucket file `k`
3. Result: same bucket file across two tables contains rows that **will join together**

Bucket Map Join (the main benefit):

- Both tables bucketed on join key with the **same number of buckets**
- Hive can JOIN bucket 0 of table A with bucket 0 of table B, bucket 1 with bucket 1, etc.
- **No full sort-merge shuffle** needed — massive speedup vs. un-bucketed JOIN

Bucketing: Hash-Based Clustering for Efficient JOINs



Bucketing DDL

```
CLUSTERED BY (user_id) INTO 32 BUCKETS STORED AS ORC;
```

Presto: Interactive SQL on Hadoop

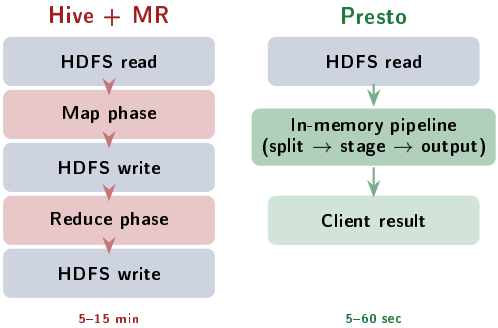
Presto: In-Memory SQL, 10× Faster than Hive+MapReduce

Why Presto Was Built (Meta, 2013)

Facebook analysts needed **interactive** (sub-second to second) SQL queries over petabytes of data on HDFS. Hive+MapReduce could not deliver this — even a simple filtered `COUNT` took minutes. **Presto**

key design decisions:

1. **No intermediate disk writes**: data flows as a pipeline through the cluster entirely in memory
2. **Vectorized execution**: processes column chunks (batches of 1,024 rows) rather than row by row
3. **Cost-based optimizer**: uses table statistics to choose optimal join order and join strategy
4. **Connector architecture**: same SQL engine for HDFS, Hive, Cassandra, MySQL, Kafka, and S3 simultaneously in one query



Student Assessment Pool

Assessment—MCQ Set A

CLO3, CLO4

Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

Q1. In Hive, where is table schema metadata (column names, types, partition info) stored?

a) HDFS NameNode
b) **Hive Metastore (relational DB such as MySQL)**
c) HBase
d) ZooKeeper

Q2. What does "schema-on-read" mean in the context of Hive?

a) Schema is validated when data is inserted into the table
b) **Schema is applied when data is queried, not when it is stored on HDFS**
c) Schema is stored separately from data in HDFS metadata
d) Schema is inferred automatically from column names in the file header

Instructor Notes

Q1: A common distractor is the NameNode — it stores HDFS block metadata, not table schemas. **Q2:**

Q3. Which Hive optimization technique physically organises data into separate HDFS sub-directories by a column's value to avoid full table scans?

(a) Bucketing (b) Indexing (c) **Partitioning** (d) Compression

Q4. Presto's primary performance advantage over Hive+MapReduce for interactive queries is:

(a) Presto supports more SQL functions than HiveQL
(b) Presto has better native integration with HBase and Cassandra
(c) **Presto executes queries entirely in memory without writing intermediate results to disk**
(d) Presto has native Python UDF support

Instructor Notes

Q3: Bucketing (a) divides *within* a partition by hash — it reduces join cost but does not skip data at the directory level. **Q4:** The key word is "intermediate results to disk." Hive+MapReduce materialises intermediate data to HDFS between every MapReduce stage. Presto pipelines everything in memory, eliminating this I/O completely for most queries.

Assessment—True / False

CLO3, CLO4

Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	Answer
1. HiveQL supports full ACID transactions on all table types, just like MySQL.	False
2. Partitioning in Hive reduces query time by eliminating irrelevant data partitions from the HDFS scan.	True
3. Hive directly executes a HiveQL query on HDFS without any intermediate translation step.	False
4. Bucketing in Hive is particularly useful for optimising JOIN operations between two large tables.	True

Assessment—Descriptive Questions

CLO3

Instructions

Answer each question in **100–150 words**. Include a diagram where indicated.

D1. Explain the **Hive architecture**: what is the role of HiveServer2, the Driver, and the Metastore? How is a HiveQL query compiled into executable jobs? Draw the component diagram.

D2. Differentiate between **partitioning** and **bucketing** in Hive. When would you choose each? Describe a scenario in Bangladesh e-commerce data where both are used together.

D3. What is "**schema-on-read**"? What are its advantages and disadvantages compared to "schema-on-write" in a traditional RDBMS such as MySQL? Give one example where schema-on-read causes a data quality problem.

Assessment—Analytical Questions

CLO4

Instructions

Answer in **200–300 words** with step-by-step reasoning and HiveQL code where applicable.

A1. Partitioning & Bucketing Design:
You have a 5 TB e-commerce transaction table with columns: (`txn_id`, `user_id`, `product_id`, `amount`, `country`, `txn_date`). Design a complete partitioning and bucketing strategy to optimise:

- Daily revenue reports filtered by `country` and `txn_date`
- JOINs between this table and a 10 M-row user profile table (keyed on `user_id`)

Write the complete `CREATE TABLE` DDL. Justify why you chose a specific number of buckets and which column(s) you partitioned on.

A2. Query Compilation to MapReduce:
Write HiveQL to find the **top 5 products by total revenue in 2023**. Then trace how Hive compiles this query into MapReduce stages: what is emitted by the Map phase, what is the shuffle key, and what does the Reduce phase compute? How would switching the execution engine from MapReduce to Tez change the execution plan?

Textbook & Reading References

Primary Textbooks for Week 7

- **[TW]** Tom White. *Hadoop: The Definitive Guide*, O'Reilly, 4th ed., 2015. **Ch. 12**
  - Ch. 12: Hive — comprehensive coverage of architecture, HiveQL, partitioning, bucketing, and UDFs
- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Ch. 5**
  - Ch. 5: Apache Hive; data warehousing on Hadoop
- **[LRU]** Leskovec, Rajaraman & Ullman. *Mining of Massive Datasets*, 3rd ed. **Ch. 2**
  - Ch. 2: SQL analogies in MapReduce; relational algebra in distributed systems

Research Articles (covered in Week 7, Part 2)

- Thusoo, A. et al. (2009). **Hive – A Warehousing Solution Over a Map-Reduce Framework**. *PVLDB 2009*.
- Thusoo, A. et al. (2010). **Hive – A Petabyte Scale Data Warehouse Using Hadoop**. *IEEE ICDE 2010*.
- Venkataraman, S. et al. (2019). **Presto: SQL on Everything**. *IEEE ICDE 2019*.

Questions & Discussion

Week 7, Part 1 | CSE 4345 Big Data Analytics

"The key insight behind Hive is not the SQL syntax — it is the Metastore: a shared, persistent catalogue that decouples the schema of your data from the files that hold it."

— Adapted from Thusoo et al., ICDE 2010

Next Lecture: Week 7, Part 2

Research Articles: Thusoo et al. (2009, 2010) on Hive architecture and petabyte-scale warehousing

Case Study: **Spotify's Hive-Based Analytics Platform** — 1 million tables, 600 billion events/day, artist royalty reporting at scale

Reading before Part 2: [TW] Ch. 12 · Thusoo et al. (2009) PVLDB paper (see references)